Predicting Earthquake Ground-Motion Directionality Differences between Nearby Sites using Machine Learning

UE24CS352A – Machine Learning · Mini-Project
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Code: https://github.com/DEEKSHAGURURAJA/earthquake-directionality-ml.

1. Problem Statement

During an earthquake, shaking at a site is strongest along one horizontal orientation (RotD100). Building codes treat this orientation as uniformly random, but observations show that nearby sites have correlated orientations. Our task (assigned problem, based on Poulos & Silva-Lopez) is to build a model *f* that takes the seismological, geological and geographical attributes of two sites and of the earthquake, and predicts the absolute difference $\Delta\theta$ (0–90°) between their orientations of maximum shaking. We compare a distance-only benchmark, ridge regression and a neural network.

2. Dataset

We use the NGA-West2 database (PEER): a flatfile with 274 attributes per record plus the two-component acceleration time series of shallow crustal earthquakes worldwide. Records from earthquakes with magnitude < 5.0 were removed (waveforms were available for 198 of the remaining records). $\Delta\theta$ is not in the flatfile, so we computed the RotD100 orientation of each record from its waveforms and paired all stations that recorded the same earthquake and are within 100 km of each other, giving 2,010 examples from 17 earthquakes. Features include earthquake source data (magnitude, mechanism, rupture size, depth), distance measures, site properties (V_{s30}, NEHRP class), ground-motion intensities (PGA, PGV, PGD) and the distance between the two sites. Numeric features were standardised (zero mean, unit variance) and categorical ones one-hot encoded. Split: 80% train, 10% validation, 10% test (1,608 / 201 / 201). Note: the reference paper limited site pairs to 5 km. PEER restricts waveform downloads to roughly 200 records per two weeks, so our dataset is much smaller than the paper's and we allowed sites up to 100 km apart; our results are therefore not directly comparable with the paper's.

3. Approach

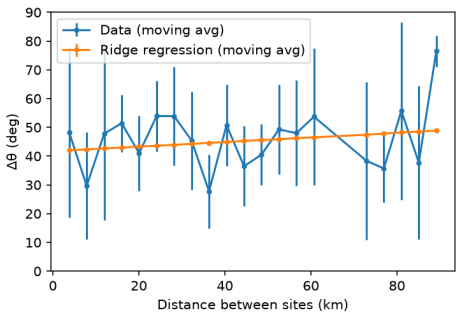
Benchmark: linear regression $\Delta\theta = c_1 + c_2 d$ using only the inter-site distance. **Ridge regression:** least squares with an L2 penalty on the coefficients; features were chosen by a backward analysis (rank each feature by the drop in validation R^2 when it is removed) followed by a forward analysis (add features in that order until validation R^2 stops improving), alternating with tuning of the regularisation parameter α on the validation set. **Neural network:** a multilayer perceptron (ReLU, Adam) using all features, with a grid search over number of layers, neurons per layer and L2 strength, selected by validation R^2. Models are compared by R^2 and mean absolute error (MAE) on the test set.

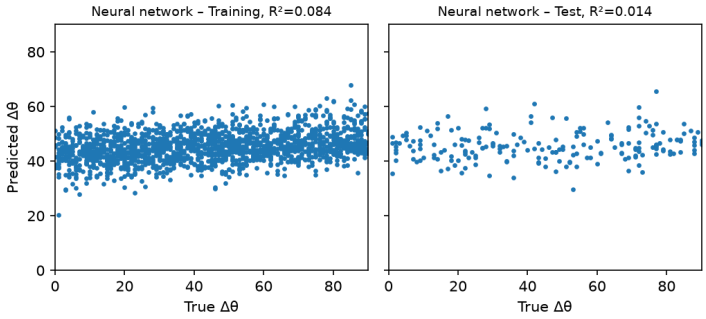
4. Implementation Overview

Python with NumPy, pandas, SciPy and scikit-learn. *waveform.py* reads the .AT2 files and finds the RotD100 orientation by passing both horizontal components through a damped single-degree-of-freedom oscillator and rotating the response through 0–179°; *pairs.py* builds the site pairs and $\Delta\theta$; *modeling.py* contains the split, preprocessing, feature selection, ridge model and neural network; *run_pipeline.py* runs everything and saves metrics, figures and models; *demo.py* predicts $\Delta\theta$ for two chosen sites. Setup and run steps are in the repository README.

5. Results

Model	R^2 (train)	R^2 (test)	MAE (test, deg)
Benchmark	0.005	0.002	23.0
Ridge regression	0.005	0.002	23.0
Neural network	0.084	0.014	22.9





6. Conclusions

The neural network had the highest test score (R^2 = 0.014, MAE = 22.9°), compared with R^2 = 0.002 for the distance-only benchmark. The neural network reached R^2 = 0.084 on training data and 0.014 on test data. All three models predicted the angle difference no better than guessing its average: the test R-squared was close to zero and the mean absolute error was about 23 degrees, close to the 22.5 degrees obtained by always predicting 45 degrees. Closer site pairs did show smaller angle differences (about 37 degrees below 10 km, against about 47 degrees at 50 to 100 km), which agrees with the reference paper, but with only 17 earthquakes and 20 pairs within 5 km there was too little short-distance data for the models to learn

from. The neural network fitted the training data better than the test data. With the full 5 km dataset of the paper, we would expect clearer results.

7. Contributions

Deeksha G: ____ . Harshitha B: ____ .

Reference: Poulos, A. and Silva-Lopez, R., Detecting and predicting earthquake ground motion directionality patterns using machine learning tools (Stanford); Ancheta et al. (2014), NGA-West2 database, Earthquake Spectra.